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Copper-Catalyzed Radical Approach to Allenyl Iodides

Yulong Song,[a] Shihua Song,[a,b] Xinyu Duan,[a,b] Xiaoyan Wu,[a,b] Feng Jiang,[a,b] Yuchen Zhang,[a,b] Junjie Fan,[a,b] Xin Huang,[a,b] Chunling Fu[a] and Shengming Ma*[a]

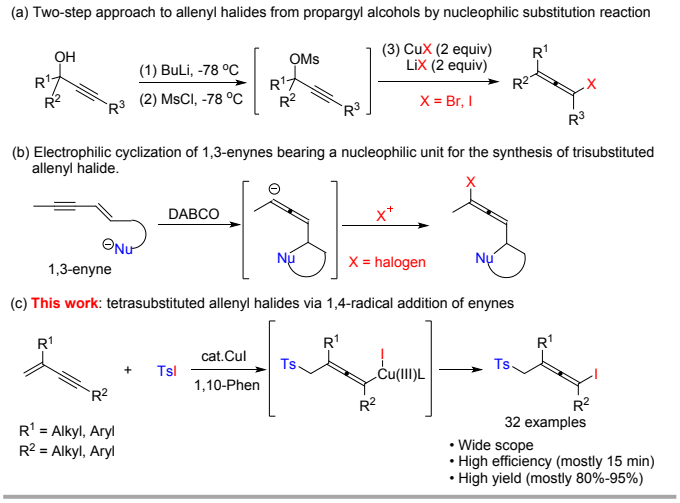
Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

A copper-catalyzed 1,4-addition reaction of sulfonyl iodides with 1,3-enynes affording various allenyl halides in high yields under mild conditions has been developed. Mechanistic studies showed that the reaction proceeds through a radical mechanism.

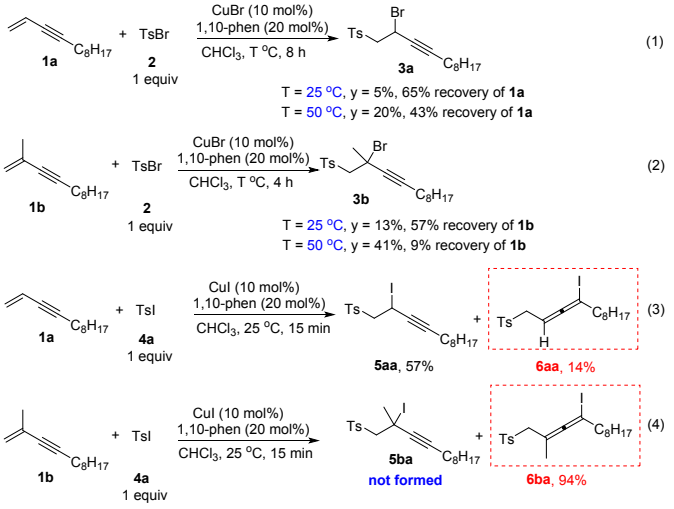
Recently much attention has been paid to the synthesis[1] and reactions[2] of allenes. Of particular interest, allenyl halides are very important building blocks in organic synthesis[3-5] as well as natural products.[6] Common approaches include (1) S_N2'-type nucleophilic substitution reaction of propargyl mesylates with two equivalents each of copper halide and lithium halide (Scheme 1 (a));[7] (2) electrophilic cyclization of 1,3-enynes bearing a nucleophilic unit for the synthesis of trisubstituted allenyl halides with a cyclic substituent (Scheme 1 (b)).[8] Thus, achieving the synthesis of allenyl halides with a high efficiency in a catalytic manner is still a challenging work. Here we wish to report that a Cu-catalyzed reaction of 1,3-enynes[9-12] with aryl sulfonyl halides[13-14] affording tetrasubstituted allenyl halides efficiently.

We commenced our study by using dodec-1-en-3-yne (**1a**) and TsBr as model substrates. After some trial and error, it was observed that propargylic bromide **3a** was obtained in 5% yield with 65% of **1a** being recovered when CuBr (10 mol %) and 1,10-phen (20 mol %) were employed for the reaction in $CHCl_3$ at 25 °C after 8 hours. By running the reaction at 50 °C, the yield of propargylic bromide **3a** was improved to 20% with 43% recovery of **1a** (Scheme 2, eq. 1). The reaction of 2-methyldodec-1-en-3-yne (**1b**) with TsBr at 25 °C also afforded propargylic bromide **3b** in 13% yield with 57% of **1b** being recovered. By increasing temperature to 50 °C, the yield of propargylic bromide **3b** was improved to 41% with 9% recovery of **1b** (Scheme 2, eq. 2). Interestingly, for the reaction of dodec-1-en-3-yne (**1a**) with TsI, in addition to 57% of pro



Scheme 1. Approaches to allenyl halides.

pargylic iodide **5aa**, 14% of allenyl iodide **6aa** was also formed (Scheme 2, eq. 3). When 2-methyldodec-1-en-3-yne (**1b**) was used, allenyl iodide **6ba** was formed exclusively with the yield of 94% (Scheme 2, eq. 4).



Scheme 2. The reaction of enynes **1a** and **1b** with TsBr or TsI.

We next investigated the reactivity of various 1,3-enynes with TsI. As shown in Table 1, various alkyl (R^1)-substituted enynes

[a] *Laboratory of Molecular Recognition and Synthesis, Department of Chemistry, Zhejiang University, Hangzhou 310027, Zhejiang, People's Republic of China.*
[b] *These authors contributed equally to this work.*
Email: masm@sioc.ac.cn.
Electronic Supplementary Information (ESI) available: [details of any supplementary information available should be included here]. See DOI: 10.1039/x0xx00000x

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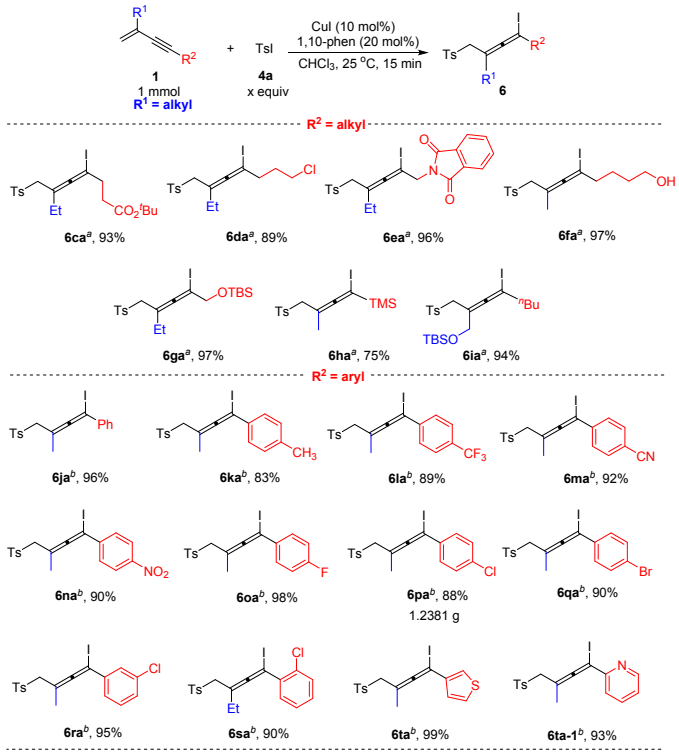
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bearing different R^2 functional groups, such as ester (**6ca**), halide (**6da**), imide (**6ea**), hydroxyl (**6fa**), and TBS ether (**6ga**) were all tolerated with the yields of 89%-97%. In addition, silyl-substituted enyne was also suitable in the reaction affording the corresponding product **6ha** in 75% yield. When R^1 was the hydroxymethyl group protected as its TBS ether, its reaction also could proceed smoothly with the yield of 94% (**6ia**).In addition to alkyl-substituted (R^2) enynes, the reaction of different aryl-substituted (R^2) enynes with TsI were also studied under the standard reaction conditions. It was found that a variety of different substitutents including electron-donating group (**6ka**) and electron-withdrawing as well as biologically or synthetically useful groups such as CF_3, CN, NO_2, F, Cl, Br (**6la-6sa**) at the *para*-, *ortho*-, or *meta*-position were well tolerated to afford the desired allenyl iodides in good to excellent yields (83%-98%). Thienyl-substituted allenyl iodide **6ta** and allenyl iodide bearing a pyridinyl group **6ta-1** could also be prepared in excellent yields. The structures of the products were further unambiguously established by X-ray crystallographic analysis of **6na** (Figure 1).[15]

Table 1 The reactions of 2-alkyl-substituted enynes with TsI.



[a] Reaction conditions: **1** (1 mmol), **4a** (1.0 equiv), CuI (10 mol%), 1,10-phen (20 mol%) in $CHCl_3$ (10 mL) at 25 °C on a 1.0 mmol scale unless otherwise noted. [b] Reaction conditions: **1** (1 mmol), **4a** (1.3 equiv), CuI (10 mol%), 1,10-phen (20 mol%) in $CHCl_3$ (10 mL) at 25 °C.

Furthermore, the substituent at the 2-position (R^1) was not limited to alkyl groups: Aryl substituent also worked (Table 2).

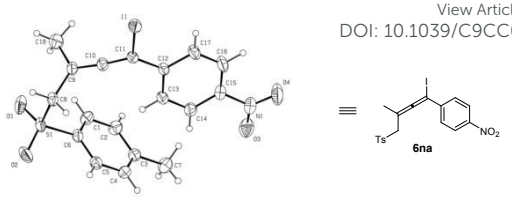
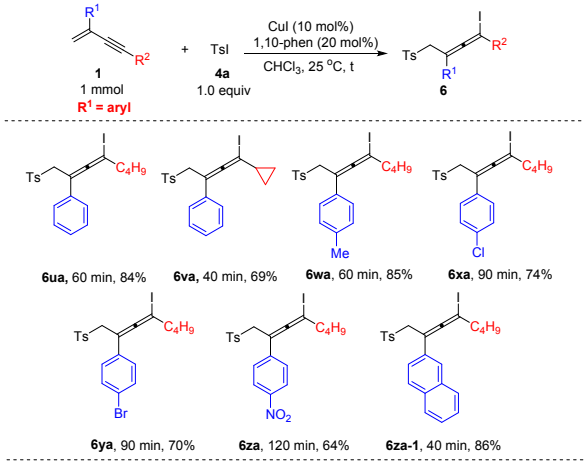


Fig. 1 ORTEP representation of **6na**.

A series of 3-aryl allenyl iodides were prepared in modest to good yields (64%-85%) with versatile substituent(s) on the aromatic rings regardless of the electron-donating or electron-withdrawing group (**6ua-6za**). It was found that the phenyl ring could also be replaced with a naphthyl ring with the yield of 86% (**6za-1**).

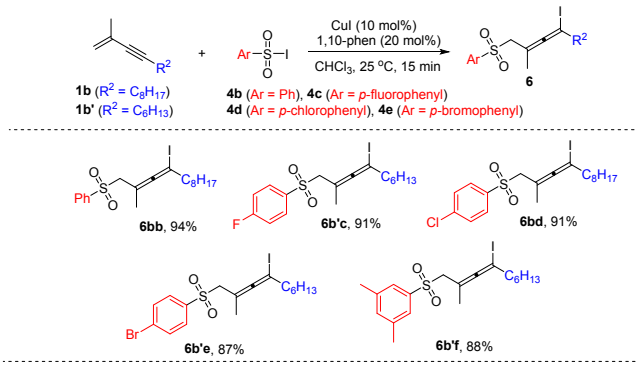
Table 2 The reactions of 2-aryl-substituted enynes with TsI.[a]



[a] Reaction conditions: **1** (1 mmol), **4a** (1.0 equiv), CuI (10 mol%), 1,10-phen (20 mol%) in $CHCl_3$ (10 mL) at 25 °C unless otherwise noted.

Phenyl or 4-halophenylsulfonyl iodides **4b**-**4e** also worked smoothly with good to excellent yields of 87%-94% (Scheme 3). The 3,5-dimethylphenyl sulfonyl iodide **4f** was also suitable in the reaction affording the corresponding product **6b'f** in 88% yield. No reaction of TsCl and **1b** was observed under the standard reaction conditions with 63% recovery of **1b** and 79% recovery of TsCl.

Table 3 Scope of free radical precursor.[a]



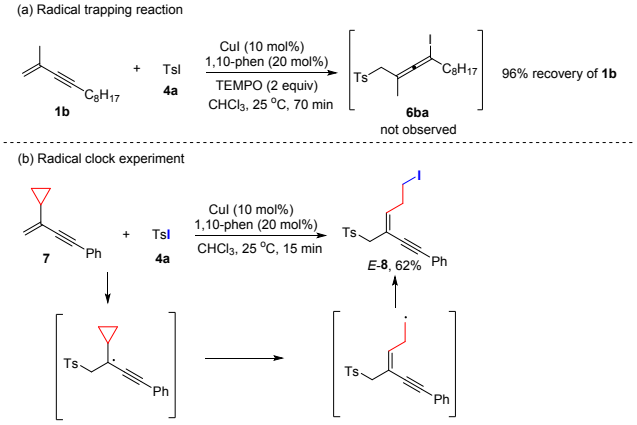
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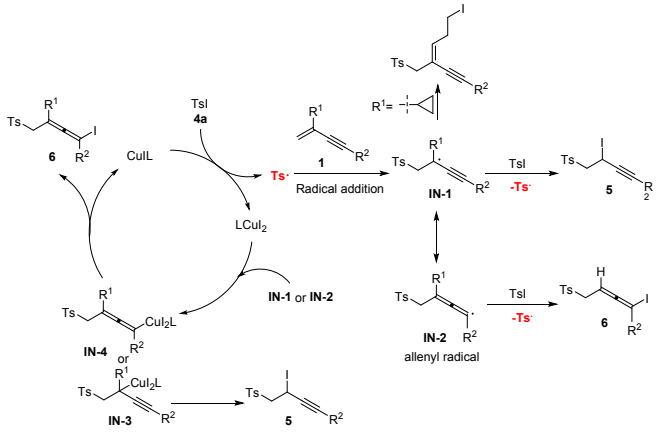
[a] Reaction conditions: **1** (1 mmol), **4** (1.0 equiv), CuI (10 mol%), 1,10-phen (20 mol%) in $CHCl_3$ (10 mL) at 25 °C on a 1.0 mmol scale.

To gain further insights into the mechanism, some control experiments have been carried out (Scheme 4). When the reaction was conducted in the presence of 2,2,6,6-tetramethy-1-piperidinyloxy (TEMPO)[16] under standard conditions, it was found that no desired product **6ba** was detected with **1b** being recovered in 96% yield. Furthermore, when enyne **7** with a cyclopropyl unit at the 2-position was treated with **4a**, the ring-opening enynyl iodide product (*E*)-**8** was produced in 62% yield, further indicating a radical mechanism. The configuration of the C=C bond in product (*E*)-**8** was determined by nOe study. The steric hindrance of Ts may determine the configuration of product (*E*)-**8**.



Scheme 4. Mechanistic studies

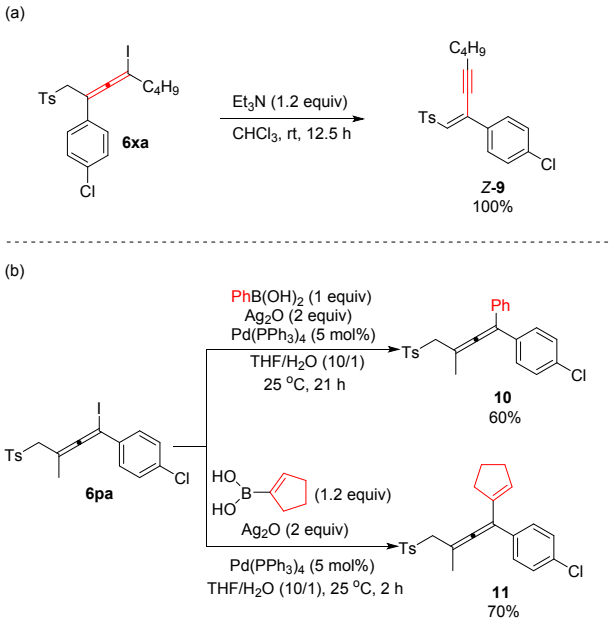
On the basis of the experimental results aforementioned, we proposed a possible mechanism (Scheme 5). Initially, the LCuI species reacts with TsI to afford the $LCuI_2$ species and the *p*-methylbenzenesulfonyl free radical via a single electron transfer process. Then the *p*-methylbenzenesulfonyl free radical attacks enyne (**1**) to produce the propargyl radical **IN-1**, which is in resonance with allenyl radical **IN-2**. These two species **IN-1** and **IN-2** would couple with $LCuI_2$ species to generate propargyl copper(III) diiodide species **IN-3** and allenyl



Scheme 5. The Proposed Mechanism

copper(III) diiodide species **IN-4**. Subsequent reductive elimination would yield the final product allene **6** or propargyl iodide **5** and regenerates LCuI species to complete this catalytic cycle. There is an alternative pathway: **IN-1** and **IN-2** would abstract the iodide from TsI to afford the final product **5** or **6** and Ts radical. The selectivity for forming **5** or **6** is determined by the steric effect of the R^1 group.

Synthetic potential of the products has also been demonstrated: (*Z*)-1,3-enyne **9** can be easily obtained in 100% yield by treating **6xa** with 1.2 equiv Et_3N (Scheme 6). The configuration of (*Z*)-**9** was determined by an NOE study. The Suzuki coupling reaction of **6pa** with aryl or alkenyl boronic acids proceeded smoothly to give the corresponding allene products **10** and **11** in 60% and 70% yield.[17]



Scheme 6. Synthetic applications.

Conclusions

In conclusion, we have developed an efficient method for the synthesis of tetrasubstituted allenyl iodides via a copper-catalyzed 1,4-radical addition reaction of 1,3-enynes. A broad range of functional groups were tolerated under mild conditions with good to excellent yields. Mechanistic studies suggested that the reaction proceeded via a radical pathway involving the resonance of propargyl radical with allenyl radical. Further studies in this area are being pursued in our laboratory.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

Financial support from the National Natural Science Foundation of China (21690063) is greatly appreciated. We thank Mr.

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Chaofan Huang in this group for reproducing the preparation of **6ga**, **6oa**, and **6wa**. Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University.

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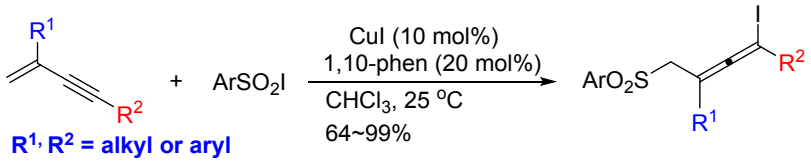
15 Crystal data for compound **6na**: $C_{18}H_{16}INO_4S$; MW=469.28; monoclinic, space group P 1 21/n 1; final R indices [I > 2\s(I)], R1=0.0358, wR2=0.0845, R indices (all data) R1=0.0530, wR2=0.0949; a = 10.4508(2) Å, b = 10.1942(2) Å, c = 17.5269(4) Å; α = 90, β = 90.2100(10), γ = 90; V= 1867.26(7) Å^3; T = 296 K; Z = 4; reflections collected/unique 23385/ 8471 (R_{int} = 0.0299); number of observations [>2\s(I)] 6153, parameters: 228. CCDC-1908167 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

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R1, R2 = alkyl or aryl

A copper-catalyzed 1,4-addition reaction of arylsulfonyl iodides with 1,3-enynes afforded various allenyl halides in high yields under mild conditions via a radical mechanism.

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